UNIT 4 - COMPLETE NOTES

Algebraic Expressions and Algebraic Formulas

What an algebraic expression is, how polynomials are classified, and the square and cube formulas that turn long multiplications into one-line answers.

What this unit covers

- **4.1** Algebraic expressions - constants, variables, terms, coefficients
- **4.2** Polynomials - types by number of terms, and degree
- **4.3** Rational and irrational expressions; adding, subtracting and multiplying expressions
- **4.4** Square formulas - $(a + b)^2$, $(a - b)^2$, $a^2 - b^2$
- **4.5** $(x + a)(x + b)$ and $(a + b + c)^2$
- **4.6** Cube formulas - $(a + b)^3$, $(a - b)^3$, $a^3 + b^3$, $a^3 - b^3$
- **4.7** Using the formulas - squares of numbers, and conditions such as $x + 1/x$

How to use these notes

Learn each formula by expanding it once yourself, then use it as a shortcut. Work every example with pen and paper before reading its solution. Yellow boxes hold the formulas and traps most often tested in board papers.

Exam focus: Board papers regularly ask for (1) the degree and type of a polynomial, (2) expanding with a square or cube formula, (3) evaluating a square such as 103^2 or 98^2 by formula, (4) finding $x^2 + 1/x^2$ or $x^3 + 1/x^3$ from $x + 1/x$, and (5) using $a^2 - b^2$ to simplify a product such as 99×101.

Based on *A Textbook of Mathematics for Grade IX*, Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar - Unit 4, starting at book page 97. Notes prepared by Shah Fahad.

1. Algebraic expressions - the building blocks

Definitions. A **constant** has one fixed value (5, -3, 1/2). A **variable** is a letter such as x, y or a that can take different values. An **algebraic expression** combines constants and variables with +, -, x, division and powers, e.g. $3x^2 - 2y + 7$.

Terms, factors and coefficients

- The parts separated by + or - signs are its **terms**: in $3x^2 - 2y + 7$ the terms are $3x^2$, -2y and 7. Each term is a product of **factors** - in $3x^2$, the factors are 3, x and x.
- The numerical factor of a term is its **coefficient**: in $3x^2$ it is 3; in -2y it is -2. A term such as x has coefficient 1.
- **Like terms** have the same variables with the same powers: $4x^2y$ and $-7x^2y$ are like; $4x^2y$ and $4xy^2$ are **unlike**.

Polynomials

Definition - Polynomial: An algebraic expression in which every variable has a whole-number power (0, 1, 2, 3, ...) and no variable appears in a denominator or under a square root. So $2x^3 - 5x + 1$ is a polynomial, but 1/x and sqrt(x) + 2 are not.

Type	Number of terms	Examples
Monomial	1	5x, $-3a^2b$, 9
Binomial	2	x + 4, 2a - 3b, $x^2 - 9$
Trinomial	3	$x^2 + 5x + 6$

Degree of a polynomial

The **degree** of a term is the sum of the powers of its variables ($3x^2y$: degree 3). The **degree of a polynomial** is the highest degree of its terms.

Example 1. For $4x^3 - 2x^2y + 5x - 8$, state the number of terms, the degree, and the coefficient of x.
Solution: Four terms; degree **3** (from $4x^3$ and $-2x^2y$); coefficient of x is **5**.

Value of an expression: Replace each variable by its given number and calculate. For x = 2, $3x^2 - 2x + 7 = 3(4) - 4 + 7 =$ **15**.

2. Rational expressions and working with expressions

Rational algebraic expression: An expression of the form p(x) / q(x), where p(x) and q(x) are polynomials and q(x) is not zero - for example (x + 2) / (x - 3), which is defined for every x except x = 3. An expression involving a square root of a variable, such as sqrt(x) + 1 or (sqrt(a) + b) / c, is an **irrational expression** (a surd expression).

Adding and subtracting - like terms only

Only like terms can be added or subtracted, and only their coefficients change. The variables and powers stay exactly as they are.

Example 2. Simplify $5x^2 + 3x - 2x^2 + 7x - 4$.
Solution: Group like terms: $(5x^2 - 2x^2) + (3x + 7x) - 4 = \mathbf{3x^2 + 10x - 4}$.

Example 3. Subtract 2a - 3b from 5a + b.
Solution: (5a + b) - (2a - 3b) = 5a + b - 2a + 3b = **3a + 4b**. Note the sign change: subtracting -3b gives +3b.

Multiplying expressions

Multiply every term of one bracket by every term of the other, add the powers when multiplying the same variable, then collect like terms.

Example 4. Multiply (x + 3)(x + 5).
Solution: = x(x + 5) + 3(x + 5) = $x^2 + 5x + 3x + 15 = \mathbf{x^2 + 8x + 15}$. This pattern becomes a formula on page 5.

Example 5. Simplify $3x(2x - 4) - 2(x^2 - 5)$.
Solution: = $6x^2 - 12x - 2x^2 + 10 = \mathbf{4x^2 - 12x + 10}$.

Watch out: x + x is 2x, but x times x is x^2 - never confuse adding with multiplying. And $3x^2 + 2x$ cannot be combined into 5 of anything: the terms are unlike.

Reducing a rational expression to its lowest form needs common factors, which is the work of Unit 5 (Factorization) and Unit 6 (Algebraic Manipulation). This unit first builds the formulas those units rely on.

3. The square formulas

An **identity** is an equation that is true for every value of its variables. The three square identities below are this unit's core - each one can be proved by simple multiplication.

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$a^2 - b^2 = (a + b)(a - b)$$

Why $(a + b)^2$ works: $(a + b)^2 = (a + b)(a + b) = a^2 + ab + ab + b^2 = a^2 + 2ab + b^2$. In words: square of a sum = square of the first + twice the product + square of the second. The middle term 2ab is positive for $(a + b)^2$ and negative for $(a - b)^2$; the two squared terms are always positive.

Example 6. Expand $(x + 4)^2$ and $(3a - 2b)^2$.
Solution: $(x + 4)^2 = x^2 + 2(x)(4) + 16 =$ **$x^2 + 8x + 16$**. $(3a - 2b)^2 = 9a^2 - 2(3a)(2b) + 4b^2 =$ **$9a^2 - 12ab + 4b^2$**.

Example 7. Expand $(2x + 1/x)^2$.
Solution: $= (2x)^2 + 2(2x)(1/x) + (1/x)^2 =$ **$4x^2 + 4 + 1/x^2$**.

Reading the formulas backwards

The same identities work in reverse: a perfect-square trinomial collapses into a squared bracket, and a difference of two squares splits into two brackets.

Example 8. Write as a square or a product: (a) $x^2 + 10x + 25$ (b) $4y^2 - 12y + 9$ (c) $25x^2 - 16$
Solution: (a) **$(x + 5)^2$**, since $2(x)(5) = 10x$. (b) **$(2y - 3)^2$**, since $2(2y)(3) = 12y$. (c) **$(5x + 4)(5x - 4)$**, by $a^2 - b^2$.

The classic trap: $(a + b)^2$ is **not** $a^2 + b^2$. The middle term 2ab is missing in that wrong answer - with $a = 3$, $b = 4$: $(3 + 4)^2 = 49$, but $9 + 16 = 25$.

4. Two more product formulas

$$(x + a)(x + b) = x^2 + (a + b)x + ab$$

The middle coefficient is the **sum** a + b and the constant term is the **product** ab. The formula also works when a or b is negative - keep their signs inside both the sum and the product.

Example 9. Expand (x + 3)(x - 7) and (y - 4)(y - 6).
Solution: (x + 3)(x - 7) = x^2 + (3 - 7)x + (3)(-7) = **x^2 - 4x - 21**. (y - 4)(y - 6) = y^2 + (-4 - 6)y + (-4)(-6) = **y^2 - 10y + 24**.

The square of a trinomial

$$(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

Read it as: the square of every term, plus twice every pair multiplied together. Signs travel with their terms, so a negative term gives negative cross terms with its neighbours.

Example 10. Expand $(x + y + z)^2$ and $(a + b - c)^2$.
Solution: $(x + y + z)^2$ = **$x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$**. $(a + b - c)^2$ = **$a^2 + b^2 + c^2 + 2ab - 2bc - 2ca$** - note c^2 itself stays positive, but both cross terms involving c turn negative.

Example 11. Expand $(2x - 3y + 1)^2$.
Solution: Squares: $4x^2$, $9y^2$, 1. Cross terms: 2(2x)(-3y) = -12xy; 2(-3y)(1) = -6y; 2(1)(2x) = 4x. Total: **$4x^2 + 9y^2 + 1 - 12xy - 6y + 4x$**.

Check yourself: A squared bracket with 2 terms expands to 3 terms; with 3 terms it expands to 6 terms (3 squares + 3 cross terms). If your expansion has a different count, a term has been dropped.

5. The cube formulas

$$(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$$

The four terms run a^3, a^2b, ab^2, b^3 with coefficients 1, 3, 3, 1. For $(a - b)^3$ the signs alternate +, -, +, -. Two shorter forms of the same identities are used constantly in condition questions:

$$(a + b)^3 = a^3 + b^3 + 3ab(a + b)$$

$$(a - b)^3 = a^3 - b^3 - 3ab(a - b)$$

Example 12. Expand $(x + 2)^3$ and $(a - 3)^3$.
Solution: $(x + 2)^3 = x^3 + 3x^2(2) + 3x(4) + 8 = \mathbf{x^3 + 6x^2 + 12x + 8}$. $(a - 3)^3 = a^3 - 9a^2 + 27a - 27$.

Sum and difference of two cubes

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

Note the middle sign: it is **opposite** in the first bracket's partner - $a^3 + b^3$ pairs with a^2 **-** $ab + b^2$, and $a^3 - b^3$ pairs with a^2 **+** $ab + b^2$. The squared terms are positive in both.

Example 13. Write as a product: (a) $x^3 + 27$ (b) $8a^3 - 1$
Solution: (a) $x^3 + 3^3 = \mathbf{(x + 3)(x^2 - 3x + 9)}$. (b) $(2a)^3 - 1^3 = \mathbf{(2a - 1)(4a^2 + 2a + 1)}$.

Example 14. Expand $(2x + 1)^3$.
Solution: $= 8x^3 + 3(4x^2)(1) + 3(2x)(1) + 1 = \mathbf{8x^3 + 12x^2 + 6x + 1}$.

Watch out: $(a + b)^3$ is not $a^3 + b^3$ - the two middle terms $3a^2b$ and $3ab^2$ are the heart of the formula, exactly as $2ab$ was for squares.

6. Using the formulas - numbers and conditions

Squares and products of ordinary numbers

Write the number as a sum or difference with a round number, then apply a formula - no column multiplication needed.

Example 15. Find (a) 103^2 (b) 98^2 (c) 99 x 101 (d) $105^2 - 95^2$

Solution: (a) $(100 + 3)^2 = 10000 + 600 + 9 =$ **10609**. (b) $(100 - 2)^2 = 10000 - 400 + 4 =$ **9604**. (c) $(100 - 1)(100 + 1) = 100^2 - 1 =$ **9999**. (d) $(105 + 95)(105 - 95) = 200 \times 10 =$ **2000**.

Condition questions: from x + 1/x to squares and cubes

The method: Square or cube both sides of the given condition and rearrange. The two results below are used again and again:

$x^2 + 1/x^2 = (x + 1/x)^2 - 2$ and $x^2 + 1/x^2 = (x - 1/x)^2 + 2$

$x^3 + 1/x^3 = (x + 1/x)^3 - 3(x + 1/x)$ and $x^3 - 1/x^3 = (x - 1/x)^3 + 3(x - 1/x)$

Example 16. If $x + 1/x = 5$, find $x^2 + 1/x^2$ and $x^3 + 1/x^3$.

Solution: $x^2 + 1/x^2 = 5^2 - 2 =$ **23**. $x^3 + 1/x^3 = 5^3 - 3(5) = 125 - 15 =$ **110**.

Example 17. If $x - 1/x = 4$, find $x^2 + 1/x^2$.

Solution: $x^2 + 1/x^2 = (x - 1/x)^2 + 2 = 16 + 2 =$ **18**. Note the sign flip: with a minus condition, you **add** 2.

Example 18. If $a + b = 7$ and $ab = 12$, find $a^2 + b^2$ and $a^3 + b^3$.

Solution: $a^2 + b^2 = (a + b)^2 - 2ab = 49 - 24 =$ **25**. $a^3 + b^3 = (a + b)^3 - 3ab(a + b) = 343 - 3(12)(7) = 343 - 252 =$ **91**.

A useful special case: If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$. Check with a = 1, b = 2, c = -3: 1 + 8 - 27 = -18, and 3(1)(2)(-3) = -18.

7. Solved examples - mixed practice

Example 19. Expand $(x - 5)^2 - (x + 5)^2$.

Solution: $= (x^2 - 10x + 25) - (x^2 + 10x + 25) =$ **$-20x$**. The squared terms cancel; only the middle terms survive.

Example 20. Simplify $(a + b)^2 + (a - b)^2$ and $(a + b)^2 - (a - b)^2$.

Solution: Sum $= (a^2 + 2ab + b^2) + (a^2 - 2ab + b^2) =$ **$2a^2 + 2b^2$**. Difference = **$4ab$**. Both results are worth remembering.

Example 21. Find 52^2 and 999^2 by formula.

Solution: $52^2 = (50 + 2)^2 = 2500 + 200 + 4 =$ **2704**. $999^2 = (1000 - 1)^2 = 1000000 - 2000 + 1 =$ **998001**.

Example 22. If $x + 1/x = 3$, find $x^4 + 1/x^4$.

Solution: First $x^2 + 1/x^2 = 3^2 - 2 = 7$. Square again: $x^4 + 1/x^4 = 7^2 - 2 =$ **47**. The same -2 step repeats at each squaring.

Example 23. Expand $(3x + 2y)^3$.

Solution: $= 27x^3 + 3(9x^2)(2y) + 3(3x)(4y^2) + 8y^3 =$ **$27x^3 + 54x^2y + 36xy^2 + 8y^3$**.

Example 24. Evaluate $(104)^3$ by formula, treating it as $(100 + 4)^3$.

Solution: $= 1000000 + 3(10000)(4) + 3(100)(16) + 64 = 1000000 + 120000 + 4800 + 64 =$ **1124864**.

Common mistakes to avoid

- Dropping the middle term: $(a + b)^2 = a^2 + b^2$ and $(a + b)^3 = a^3 + b^3$ are both wrong.
- Making the squared terms negative in $(a - b)^2$: a^2 and b^2 stay positive; only $2ab$ is negative.
- Using -2 with a minus condition: $x^2 + 1/x^2 = (x - 1/x)^2$ **+ 2**, not - 2.
- Wrong middle sign in the cube pairings: $a^3 + b^3$ goes with $a^2 - ab + b^2$.
- In $(x + a)(x + b)$, swapping sum and product: the x-coefficient is $a + b$, the constant is ab.

8. Test yourself - MCQs and short questions

Multiple-choice questions

1. The degree of $5x^2y - 3x + 7$ is: (a) 2 (b) 3 (c) 4 (d) 1
   Answer: (b) - $5x^2y$ has degree 3.
2. $(a - b)^2$ equals: (a) $a^2 - b^2$ (b) $a^2 + 2ab + b^2$ (c) $a^2 - 2ab + b^2$ (d) $a^2 - 2ab - b^2$
   Answer: (c)
3. $(x + 6)(x - 6)$ equals: (a) $x^2 + 36$ (b) $x^2 - 36$ (c) $x^2 - 12x - 36$ (d) $(x - 6)^2$
   Answer: (b) - difference of two squares.
4. 102^2 equals: (a) 10004 (b) 10404 (c) 10204 (d) 10040
   Answer: (b) - $(100 + 2)^2 = 10000 + 400 + 4$.
5. If $x + 1/x = 4$, then $x^2 + 1/x^2$ equals: (a) 16 (b) 18 (c) 14 (d) 12
   Answer: (c) - $4^2 - 2 = 14$.
6. $a^3 - b^3$ equals: (a) $(a - b)(a^2 - ab + b^2)$ (b) $(a - b)(a^2 + ab + b^2)$ (c) $(a + b)(a^2 + ab + b^2)$ (d) $(a - b)^3$
   Answer: (b)
7. $(x + 2)(x + 9)$ equals: (a) $x^2 + 11x + 18$ (b) $x^2 + 18x + 11$ (c) $x^2 + 11x + 11$ (d) $x^2 + 7x + 18$
   Answer: (a) - sum 11, product 18.
8. The expression $1/x + 3$ is: (a) a polynomial (b) not a polynomial, because x is in a denominator (c) a monomial (d) a trinomial
   Answer: (b)

Short questions for written practice

1. State the number of terms, the degree and the coefficient of x^2 in $3x^3 - 5x^2 + x - 9$.
2. Expand: (a) $(5x - 3)^2$ (b) $(x + y - z)^2$ (c) $(a - 2)^3$
3. Find by formula: (a) 97^2 (b) 201×199 (c) 1003^2
4. If $x - 1/x = 6$, find $x^2 + 1/x^2$ and $x^3 - 1/x^3$. (Answers: 38; 234)
5. If $a + b = 9$ and $a - b = 5$, find ab and $a^2 + b^2$. (Answers: 14; 53)
6. Write as a product: (a) $49x^2 - 36y^2$ (b) $x^3 - 64$ (Answers: $(7x + 6y)(7x - 6y)$; $(x - 4)(x^2 + 4x + 16)$)

One-page revision: Terms join to make expressions; polynomials have whole-number powers and are named monomial / binomial / trinomial, with degree = highest sum of powers in a term. Formulas: $(a +/- b)^2 = a^2 +/- 2ab + b^2$; $a^2 - b^2 = (a + b)(a - b)$; $(x + a)(x + b) = x^2 + (a + b)x + ab$; $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$; $(a +/- b)^3 = a^3 +/- 3a^2b + 3ab^2 +/- b^3$; $a^3 +/- b^3$ pair with $a^2 -/+ ab + b^2$. Conditions: square and subtract (or add) 2; cube and subtract (or add) 3 times the original.

End of Unit 4 notes. Next unit: Unit 5 - Factorization.